# Understanding the Existing 8-Bit SAR ADC Board

Beginner Study Notes—Part 1

ADC Board Educational Project

October 2026

*Purpose: Understand the basic operating principle before reading the schematic or redesigning the board.*

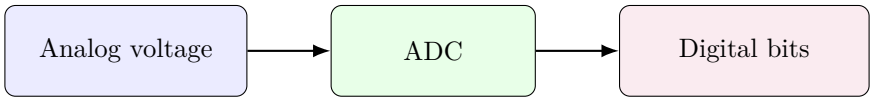
## 1 What is the goal of the project?

The professor's project brief asks for an educational **8-bit successive-approximation ADC board**. The starting point is an existing board, which must be reengineered and adapted. The brief requests (1) an external adjustable clock input named `CLK_IN`, (2) an analog input that can be provided by an external waveform generator, and (3) measurement/test pins for relevant internal signals. Student instructions and a final report are also required.[1]

**For now, the goal is only to understand the existing board.** We are not redesigning it yet.

## 2 What is an ADC?

ADC means **Analog-to-Digital Converter**. It converts an analog electrical quantity, usually an input voltage, into a digital number.




**Why is it needed?** Real-world sensors can produce continuously varying voltages. Digital logic and computers work with numbers encoded in 0s and 1s. An ADC links the two.

## 3 Analog voltage versus digital information

### 3.1 Analog

An analog voltage can take many intermediate values within an allowed interval: for example, 1.00 V, 1.25 V, 2.97 V, or 3.00 V. A potentiometer can provide an adjustable voltage.

### 3.2 Digital

Digital logic uses two states: **0 (LOW)** and **1 (HIGH)**. One binary digit is a **bit**. Eight bits together might look like `10011001`.

[1] Source: *ELT10250_ADC-Board.pdf*, page 1.

### 3.3 Important misunderstanding: does the ADC know 3 V already?

If we set a potentiometer to 3 V, the input pin indeed receives a physical 3 V signal. However, no binary number "3" arrives along the wire. The ADC must use internal electronics to determine which digital code corresponds to that physical voltage.

**Remember this**

**Receiving a voltage is not the same as already knowing its numerical digital value.**

## 4 What does 8-bit mean?

With $N$ bits there are $2^N$ distinct codes. Thus an 8-bit ADC has

$$2^8 = 256 \text{ possible codes}, \qquad 0, 1, 2, \ldots, 255.$$

For a nominal 0–5 V input range, one ideal code interval is

$$\Delta V = \frac{5 \text{ V}}{256} = 0.01953125 \text{ V} \approx 19.5 \text{ mV}.$$

This smallest nominal interval is often called **one LSB** (least significant bit). The board documentation describes approximately 19.5 mV resolution.[2]

### 4.1 How to read eight bits

Each bit has a positional weight:

$$\underbrace{128 \quad 64 \quad 32 \quad 16 \quad 8 \quad 4 \quad 2 \quad 1}_{\text{weights from left to right}}.$$

For example,

$$\texttt{10011001}_2 = 128 + 16 + 8 + 1 = 153_{10}.$$

An input of 3 V would ideally fall in code 153 under the common $\lfloor 256 V_{\text{in}}/5 \rfloor$ convention (clipped to 255). This is an *ideal model*; the physical board can differ slightly due to resistor tolerance, comparator offset and boundary behavior.

## 5 What is SAR and why is it needed?

SAR means **Successive Approximation Register**. It is one method of converting an unknown voltage into a binary code.

Instead of testing every possible code in order, SAR uses a **binary-search strategy**: begin near the middle, find out whether the unknown voltage is higher or lower, and then search only the relevant half. It repeats this process from the most significant bit (MSB) to the least significant bit (LSB).

**Remember this**

Each comparison provides one yes/no decision. One comparison is not enough to distinguish 256 possible codes; eight binary decisions can distinguish $2^8 = 256$ possibilities.

[2] Source: *discrete-adc-DOC.pdf*, PDF page 7, "What Is an ADC and Why Does It Matter?"

### 5.1 Why does the trial start at 2.5 V?

For a 0–5 V reference range, the first midpoint is 2.5 V. This is generated by the internal digital-to-analog converter (DAC) when the SAR tests the most significant bit (binary `10000000`). It is the first *trial voltage*, not an externally supplied second input.

### 5.2 Does SAR wait for exactly equal voltages?

No. The converter is designed to determine the correct *range/code*, not to find a perfectly equal analog voltage. At exact equality, a real comparator's decision may depend on offset and noise. The conversion still proceeds through the planned bit decisions.

## 6 Worked SAR example: input voltage = 3 V

Assume the input remains at 3 V during one conversion and the nominal reference is 5 V. A simplified ideal binary-search sequence is:

**Worked example**

| Step | Trial voltage | Is 3 V ≥ trial? | Bit decision |
|---|---|---|---|
| 1 | 2.500000 V | Yes | Keep bit 7 = 1 |
| 2 | 3.750000 V | No | Clear bit 6 = 0 |
| 3 | 3.125000 V | No | Clear bit 5 = 0 |
| 4 | 2.812500 V | Yes | Keep bit 4 = 1 |
| 5 | 2.968750 V | Yes | Keep bit 3 = 1 |
| 6 | 3.046875 V | No | Clear bit 2 = 0 |
| 7 | 3.0078125 V | No | Clear bit 1 = 0 |
| 8 | 2.98828125 V | Yes | Keep bit 0 = 1 |

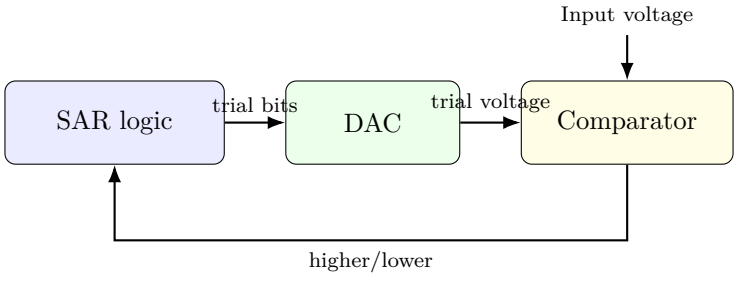
The final binary code is `10011001`, or decimal 153.

Note how the trial voltages move *up and down*. The SAR does **not** test 1 V, then 2 V, then 3 V. Each decision removes half of the remaining possibilities.

## 7 Why is there a DAC inside an ADC?

DAC means **Digital-to-Analog Converter**: it converts a binary code into an analog voltage. This is the opposite direction from the ADC as a whole.

The SAR logic only handles binary decisions. The comparator, however, must compare two *actual voltages*. So SAR sends a trial digital code to the DAC, and the DAC produces a trial voltage such as 2.5 V or 3.75 V.



The existing board uses a resistor-ladder DAC rather than a dedicated DAC chip. The documentation calls this an **R–2R ladder** and explains that it is driven by logic gates.[3]

# 8 What is a comparator?

A comparator is a circuit that indicates which of its two input voltages is greater.

For the existing board, the **LM393** compares the analog input $V_{\text{in}}$ with the DAC trial output. Conceptually:

$$V_{\text{in}} > V_{\text{DAC}} \quad \Rightarrow \quad \text{HIGH decision},$$

$$V_{\text{in}} < V_{\text{DAC}} \quad \Rightarrow \quad \text{LOW decision}.$$

The LM393 has an open-collector output, so the circuit also uses a pull-up resistor. We will learn the pin-level details later.[4]

For example: if the unknown input is 3 V and the DAC generates 2.5 V, the comparator indicates that the input is higher. It does **not** directly announce "3 V".

# 9 Two different kinds of repetition

## 9.1 Repetition inside one conversion

The ADC performs **eight comparisons** to decide its eight output bits. All eight comparisons together constitute **one complete conversion**.

## 9.2 Repetition of complete conversions

After finishing a conversion, the existing board starts another one. This is useful because the potentiometer voltage might change. The documentation says the LED result updates roughly **ten times per second**.[5]

Even if the input remains at 3 V, the board keeps measuring it. The result generally stays the same until the input changes.

**Remember this**

**Eight comparisons** ⟶ one voltage measurement.
**Repeated eight-comparison cycles** ⟶ new measurements over time.

# 10 What the existing board contains (high-level overview)

According to the board documentation, its important functions and example ICs are:

[3] Source: *discrete-adc-DOC.pdf*, PDF pages 8, 15–16.
[4] Source: *discrete-adc-DOC.pdf*, PDF pages 20–21.
[5] Source: *discrete-adc-DOC.pdf*, PDF page 8, Section 2.2.

| Block | Example device | Job |
|---|---|---|
| Power | LM7805 | Regulates the board's main logic supply to 5 V. |
| Clock | 74HC14 | Provides the timing pulses that advance the sequence. |
| SAR sequencing | CD4017 | Steps through the conversion phases. |
| Bit storage | 74HC74 | Holds bit decisions. |
| Trial voltage generation | 74HC32 and R–2R network | Produces DAC trial voltages. |
| Voltage comparison | LM393 | Compares analog input against DAC output. |
| Output/LED display | 74HC574 | Holds the completed 8-bit result for eight LEDs. |

Source: *discrete-adc-DOC.pdf*, PDF pages 8 and 15–22. These are functions to recognize for now, not a substitute for checking the actual schematic.

## 11 Questions we have answered

1. **If we supply 3 V, does 3 V really reach the ADC?** Yes. The ADC receives the real voltage, but must convert it into a numerical code.
2. **Does SAR compare 1 V, 2 V, 3 V in order?** No. It uses midpoint trials to eliminate half the possibilities per comparison.
3. **Is the trial voltage always constant?** The first trial is nominally 2.5 V; later trials change based on earlier decisions.
4. **What if input equals the trial voltage?** The converter does not stop; exact equality is not a special completion signal.
5. **Why use a DAC?** The DAC turns SAR's digital trial code into an analog voltage for comparison.
6. **Why measure repeatedly?** To update the display if the input changes.